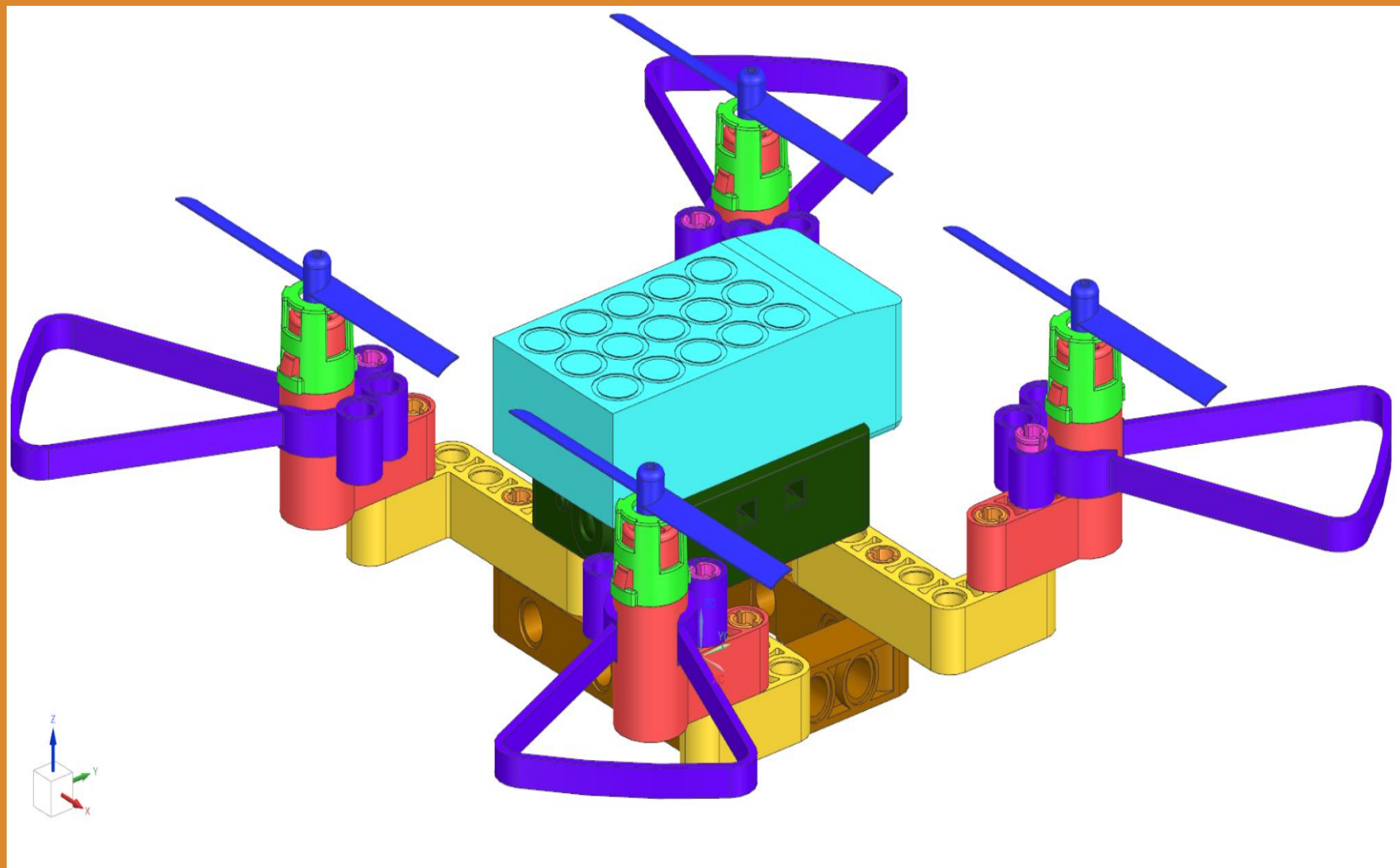


AME 308 FINAL PROJECT

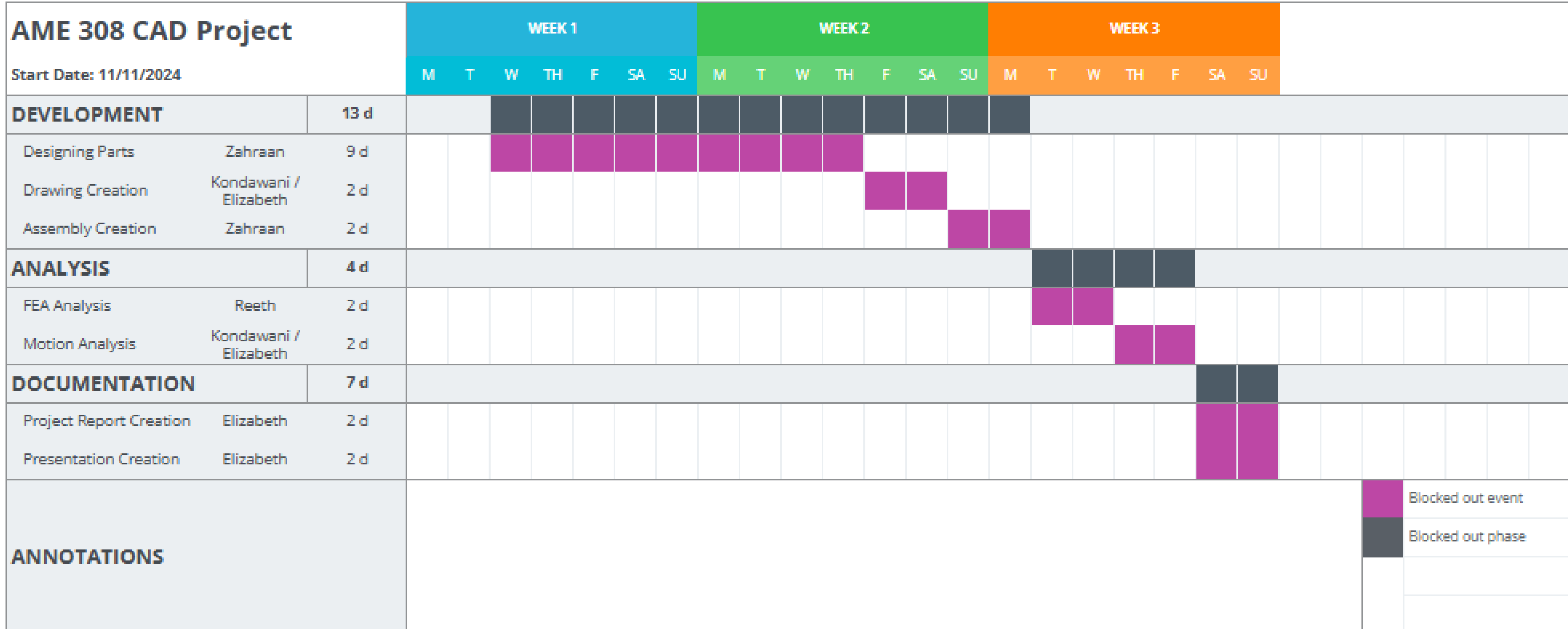


By: Elizabeth Finley, Kondwani
Phiri, Reeth Kawad, and
Zahraan Raheel

Our Drone

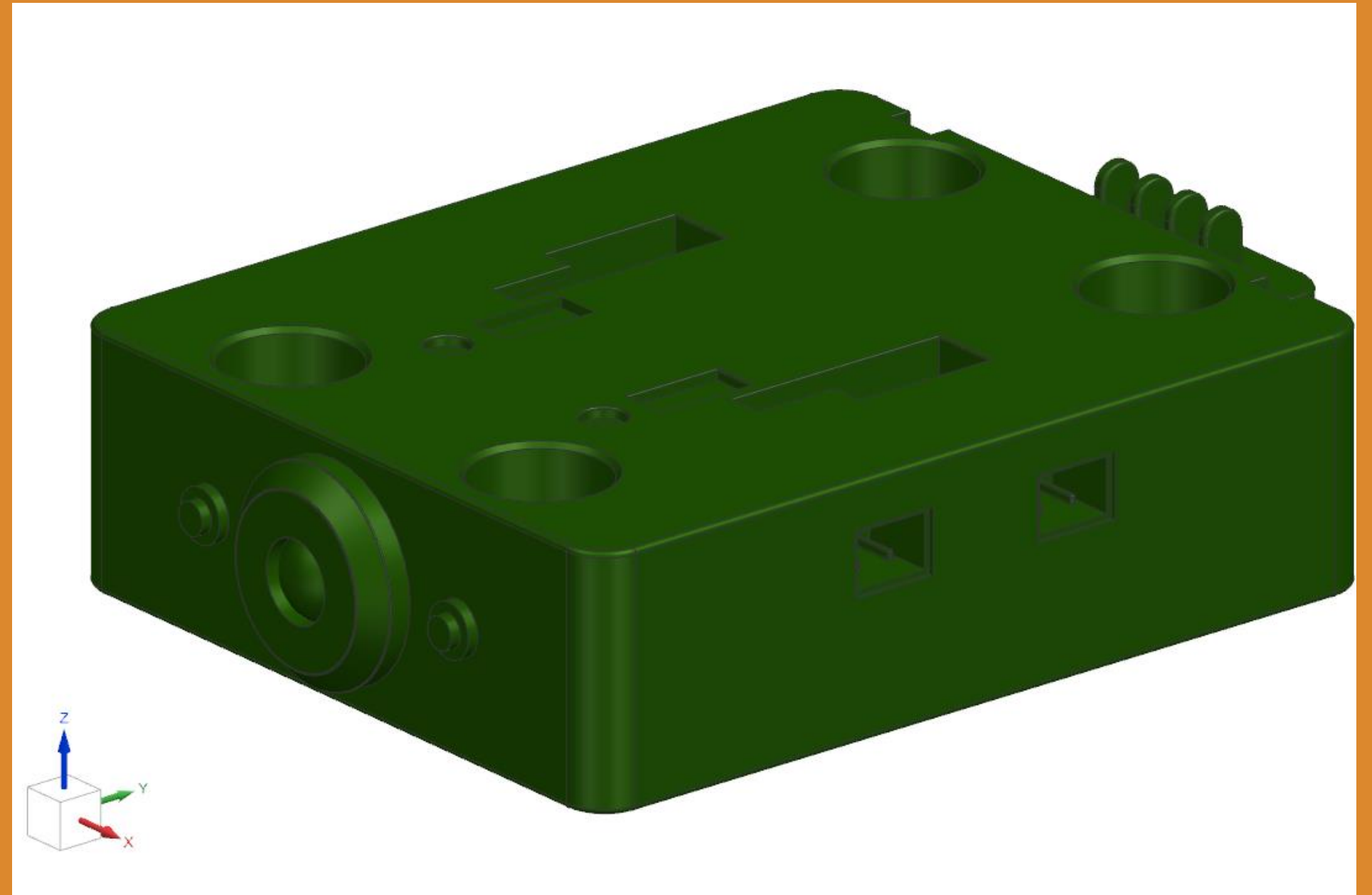
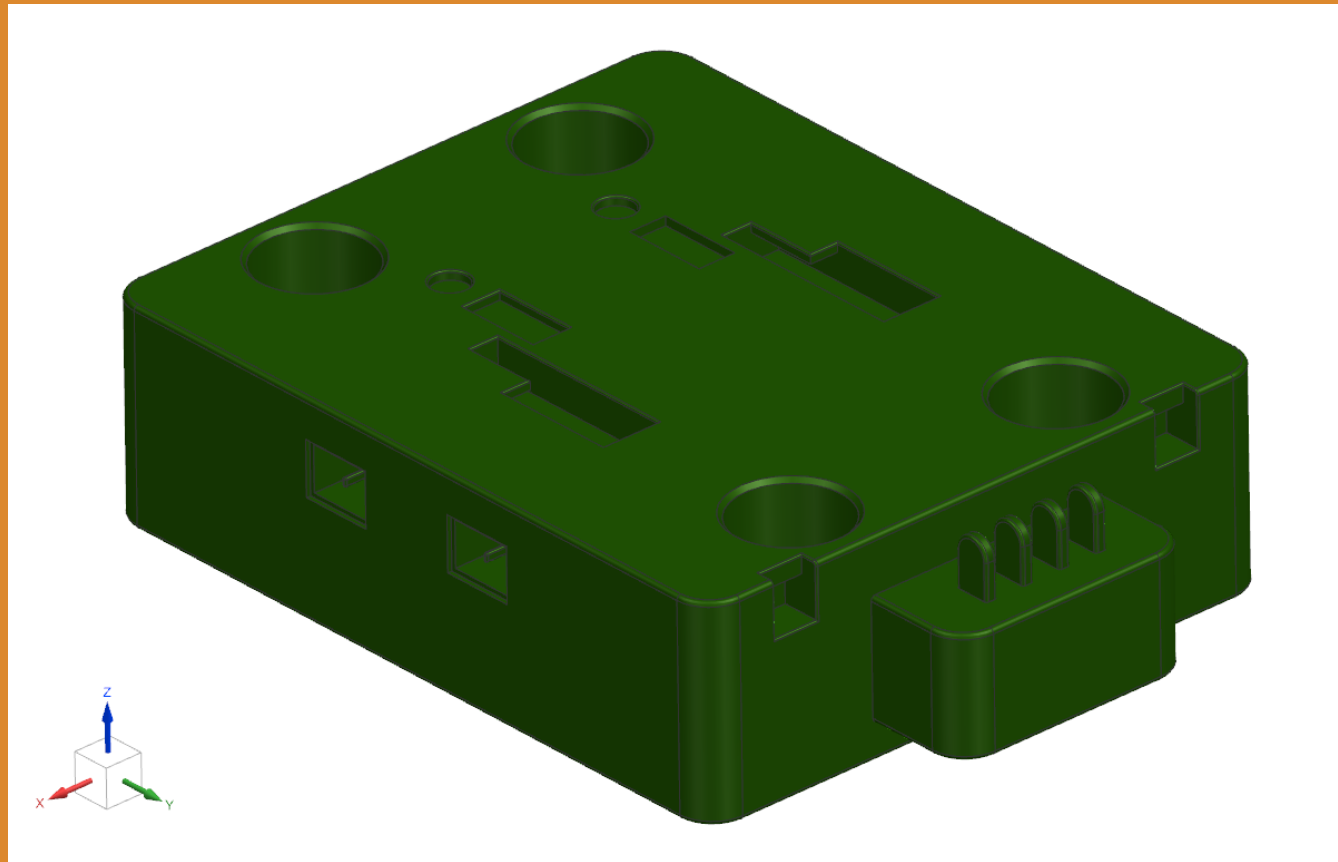


Gantt Chart

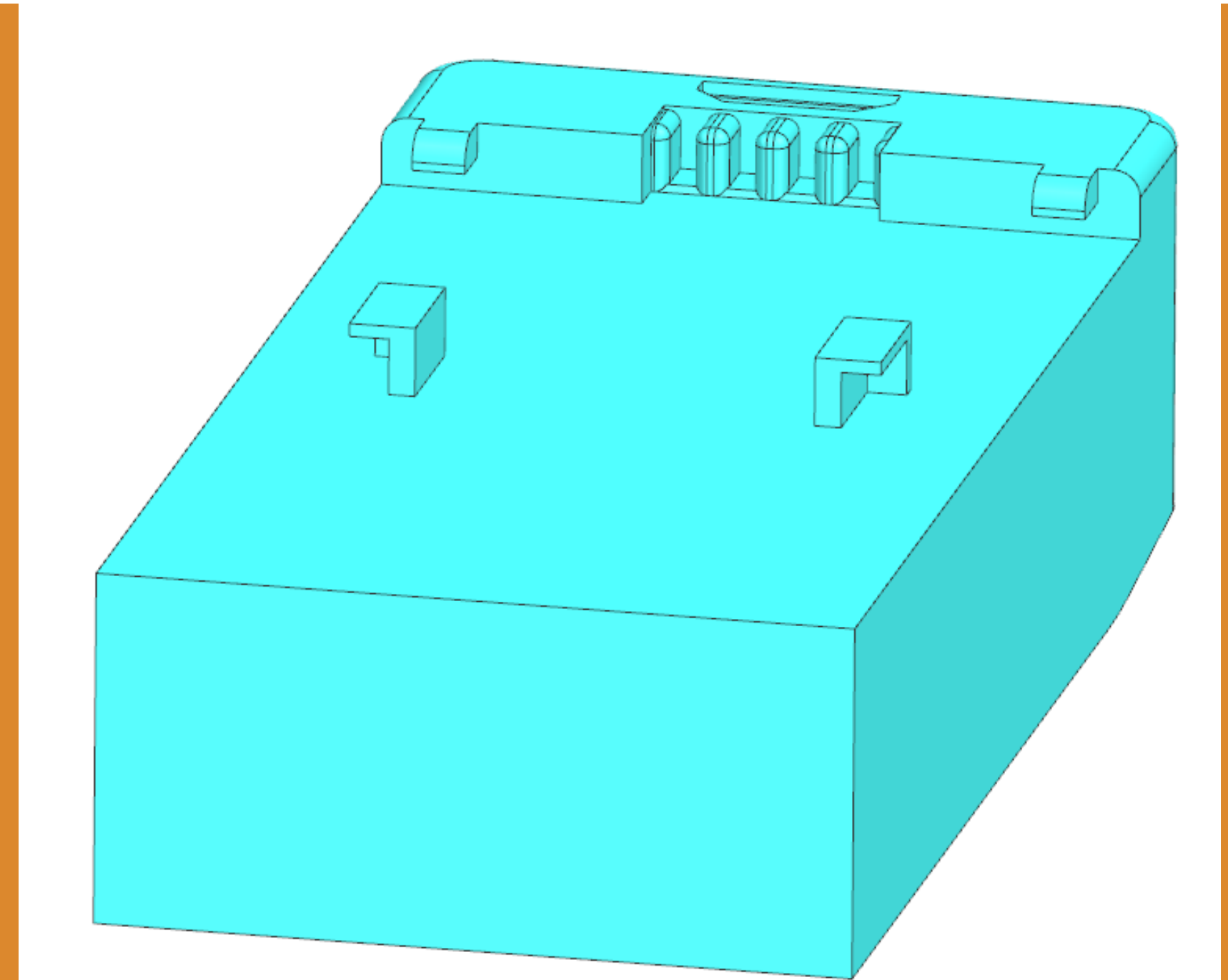
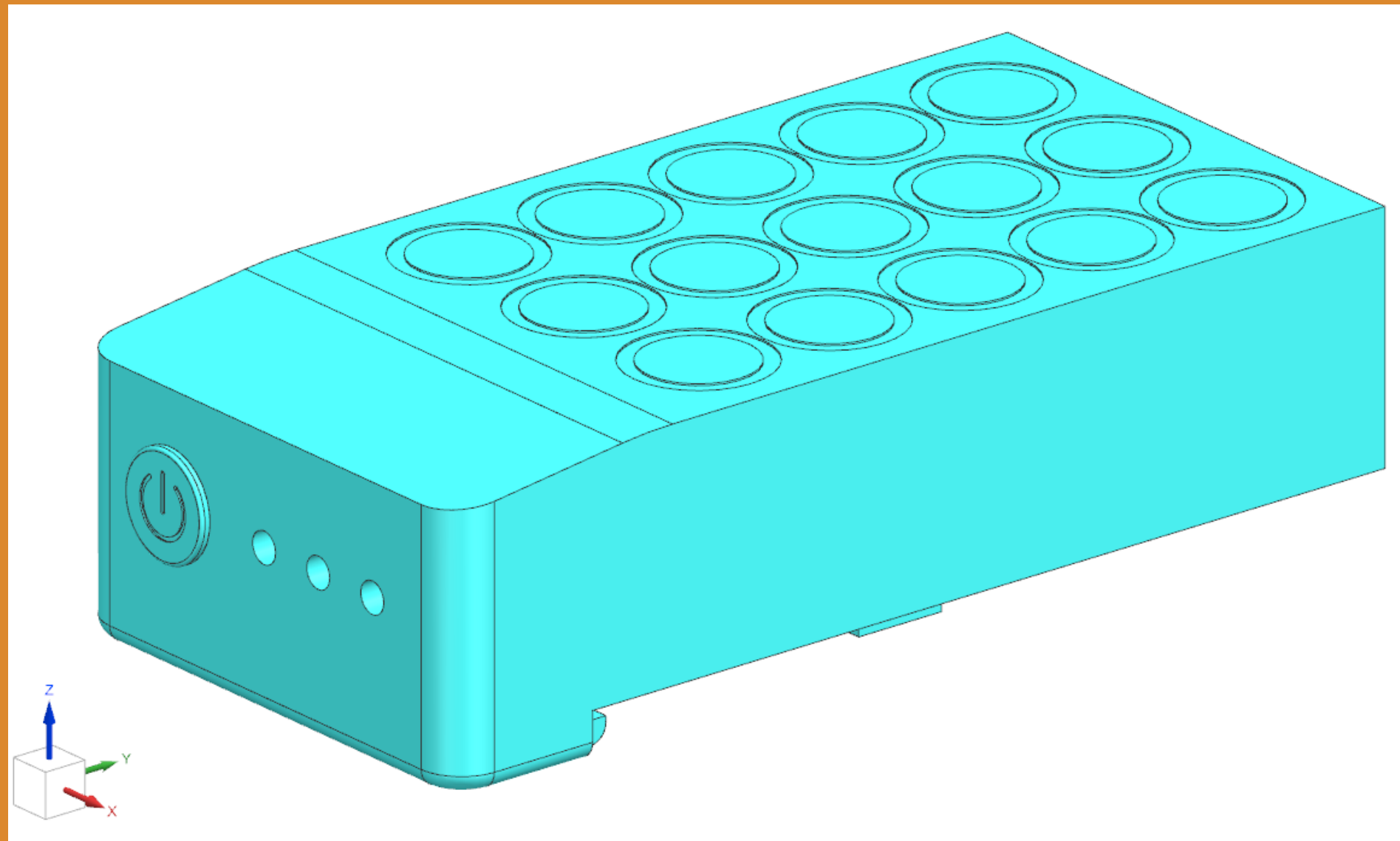


Our Parts

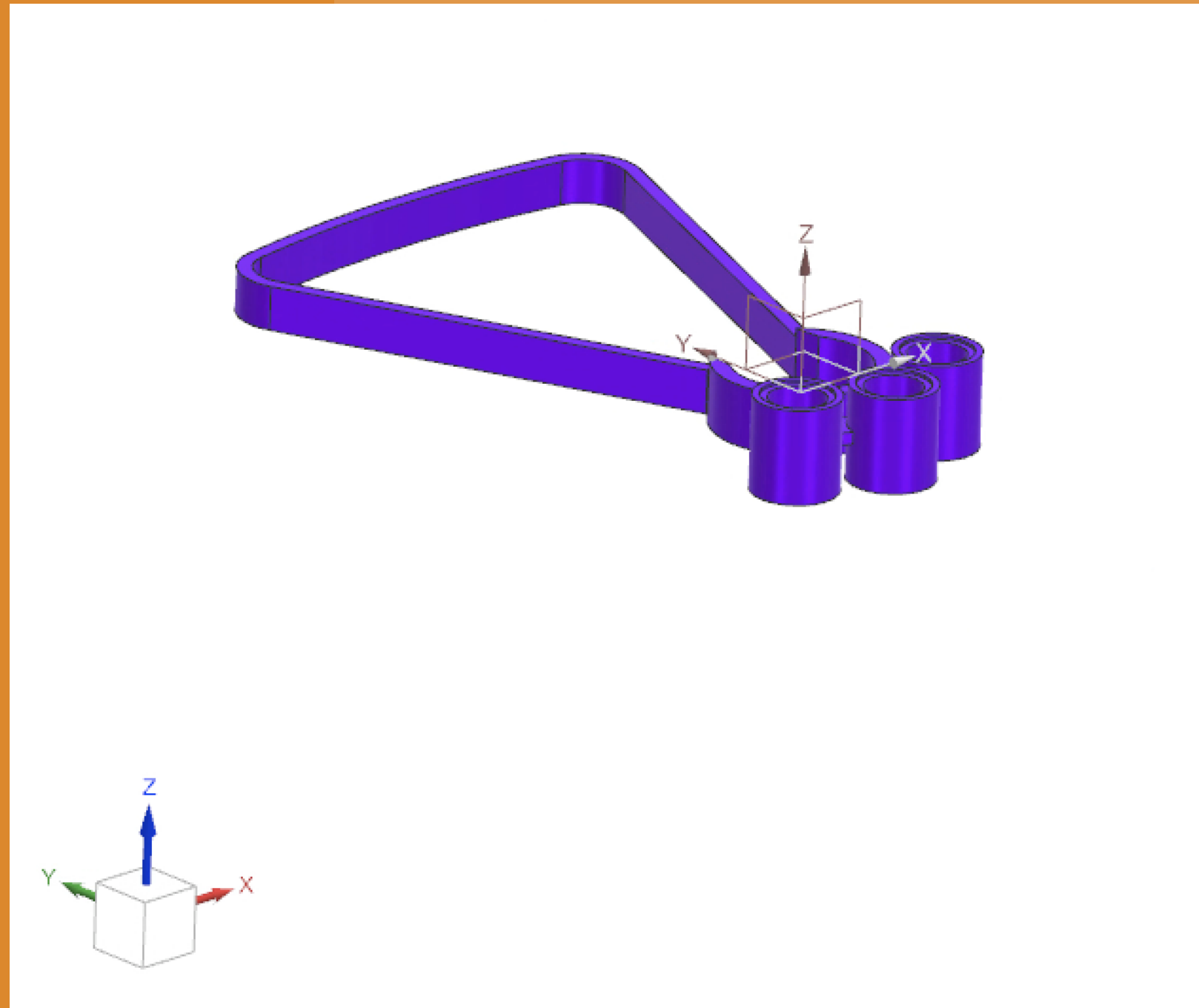
PCB Body



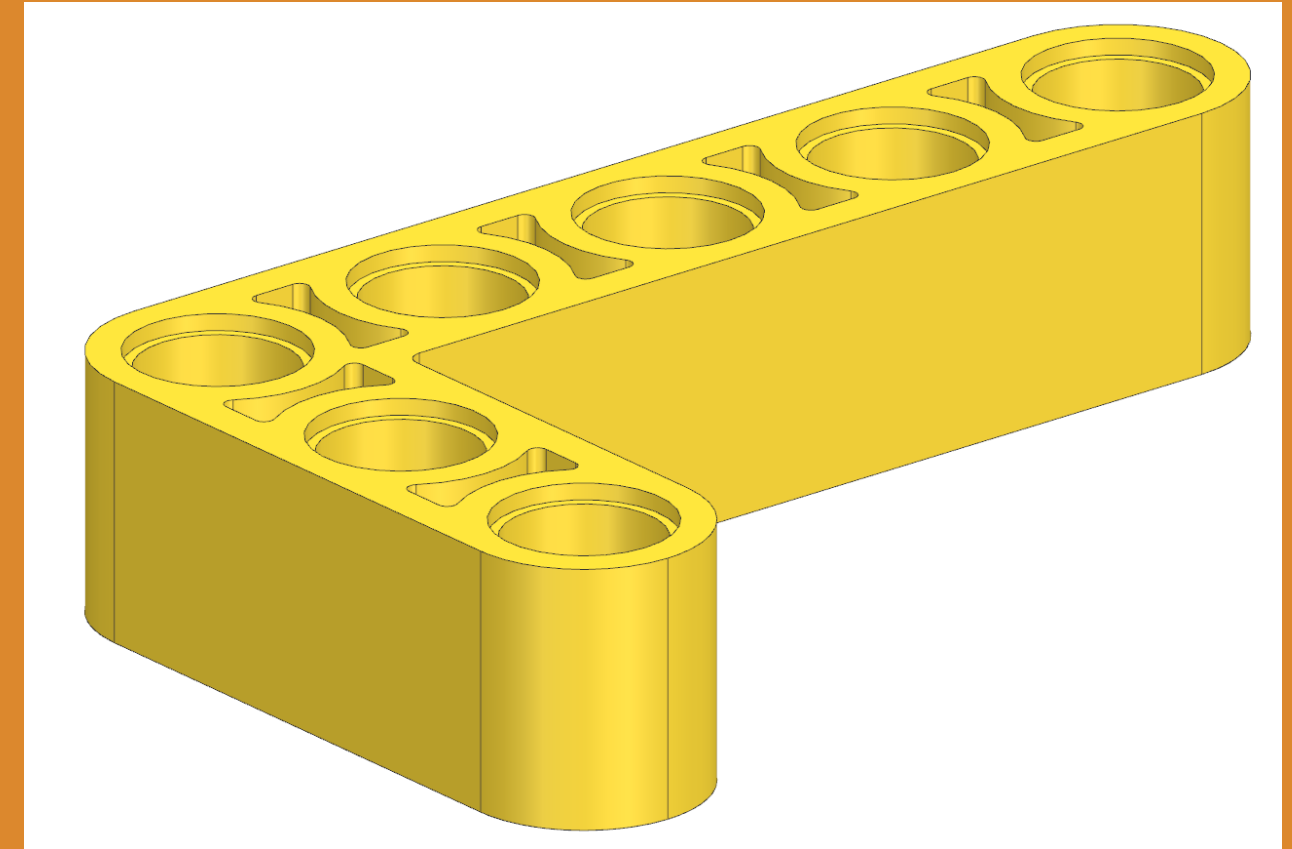
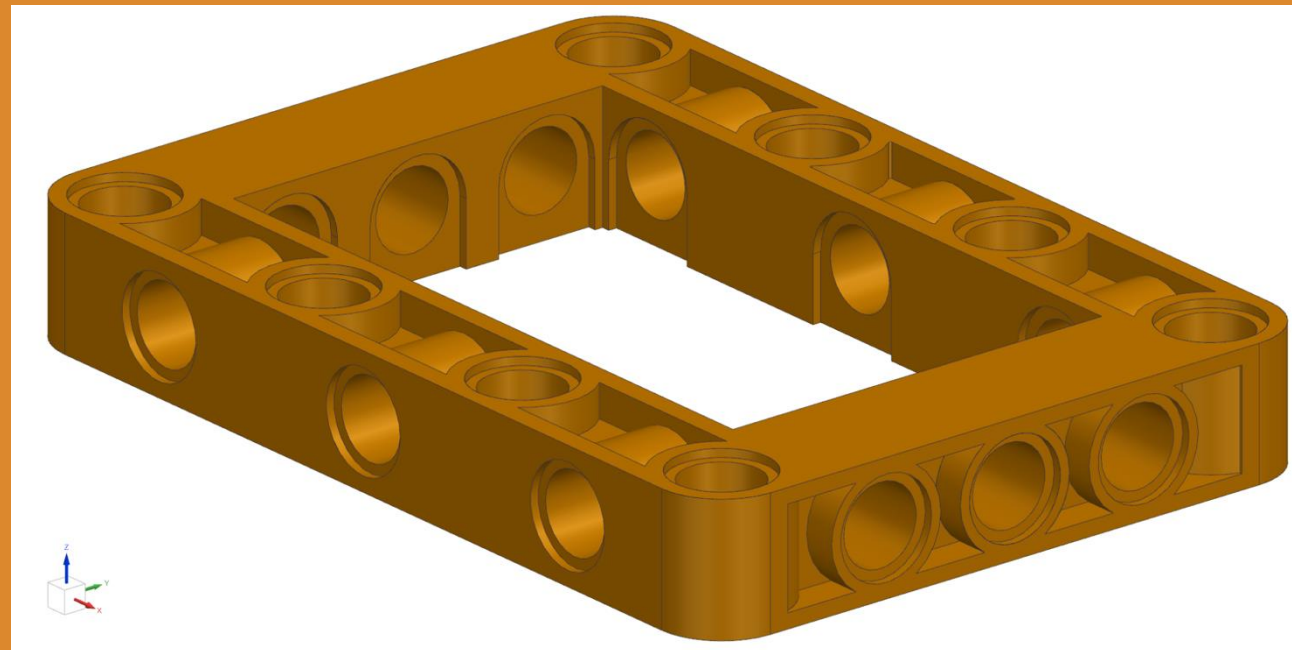
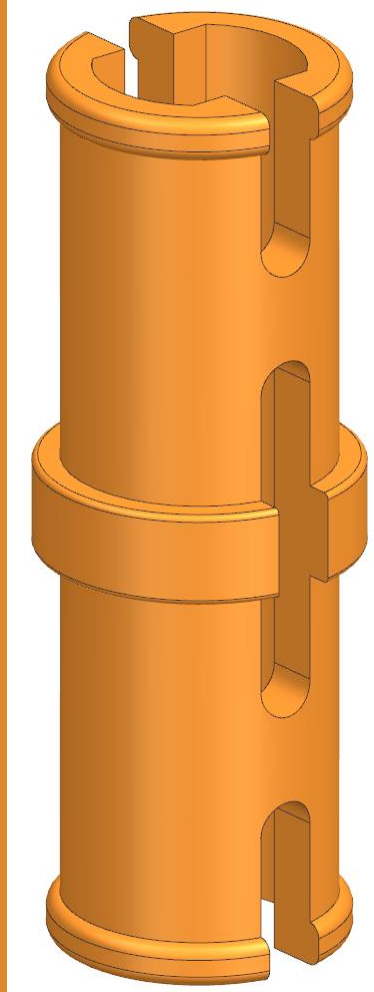
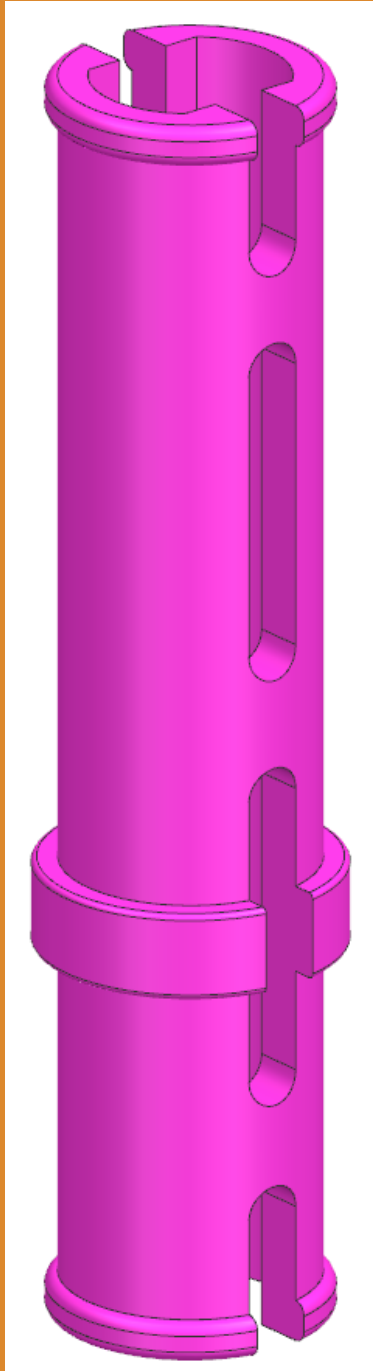
Battery Pack



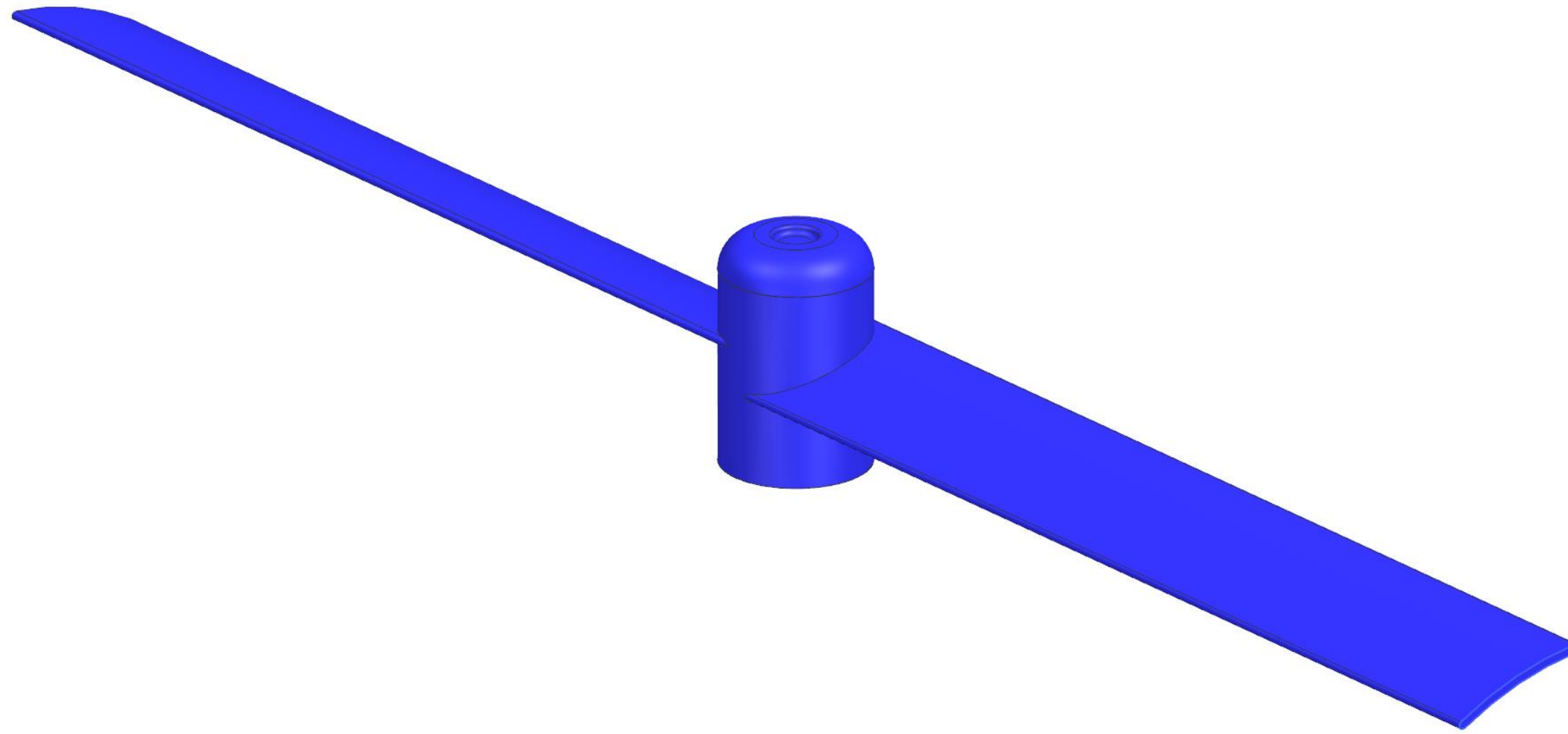
Guard Rail



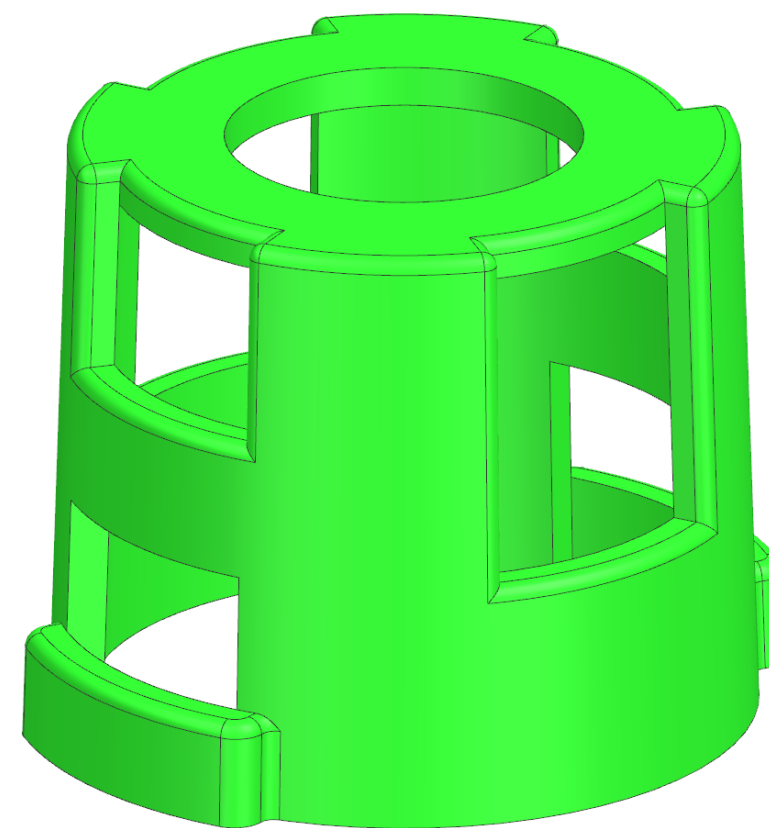
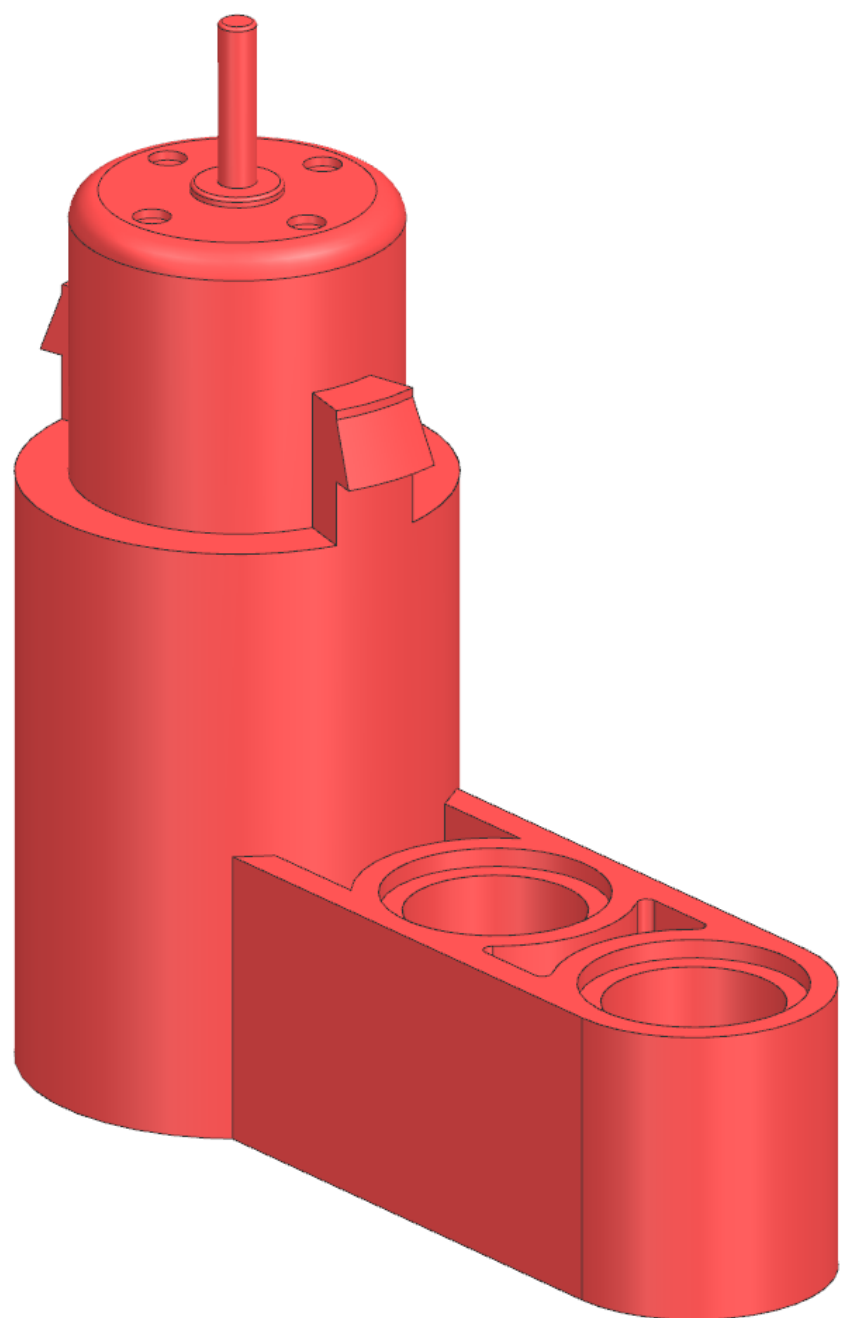
Lego Blocks



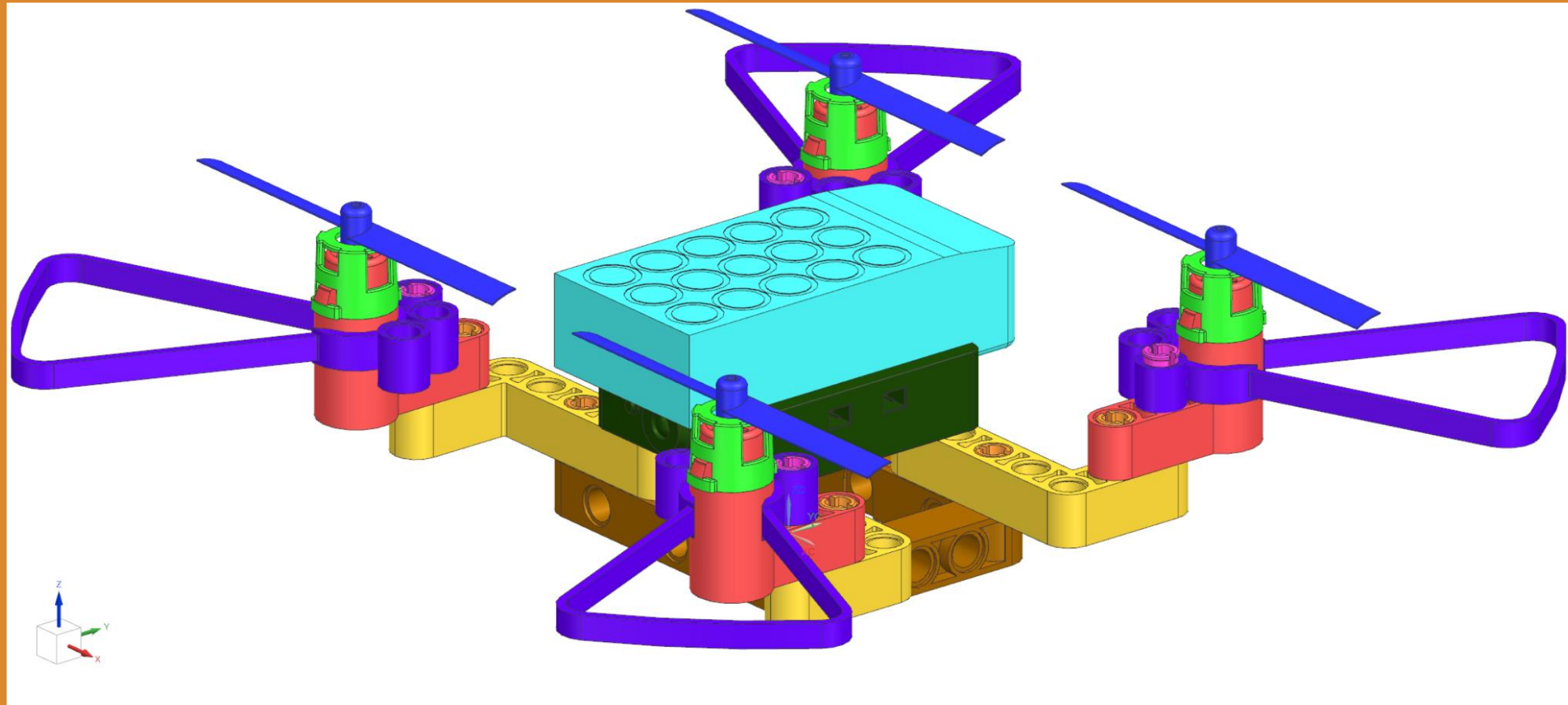
Propeller



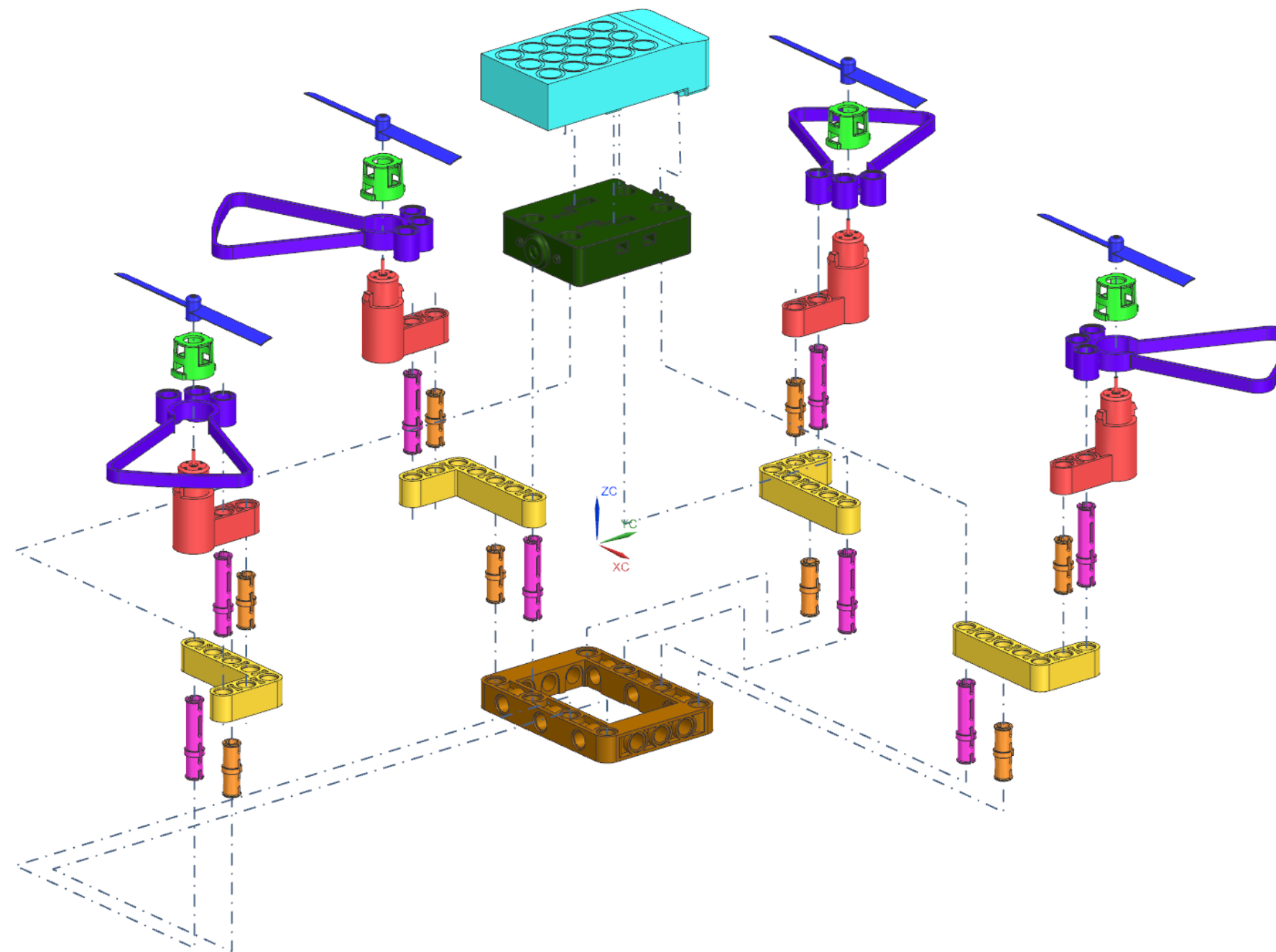
Motor



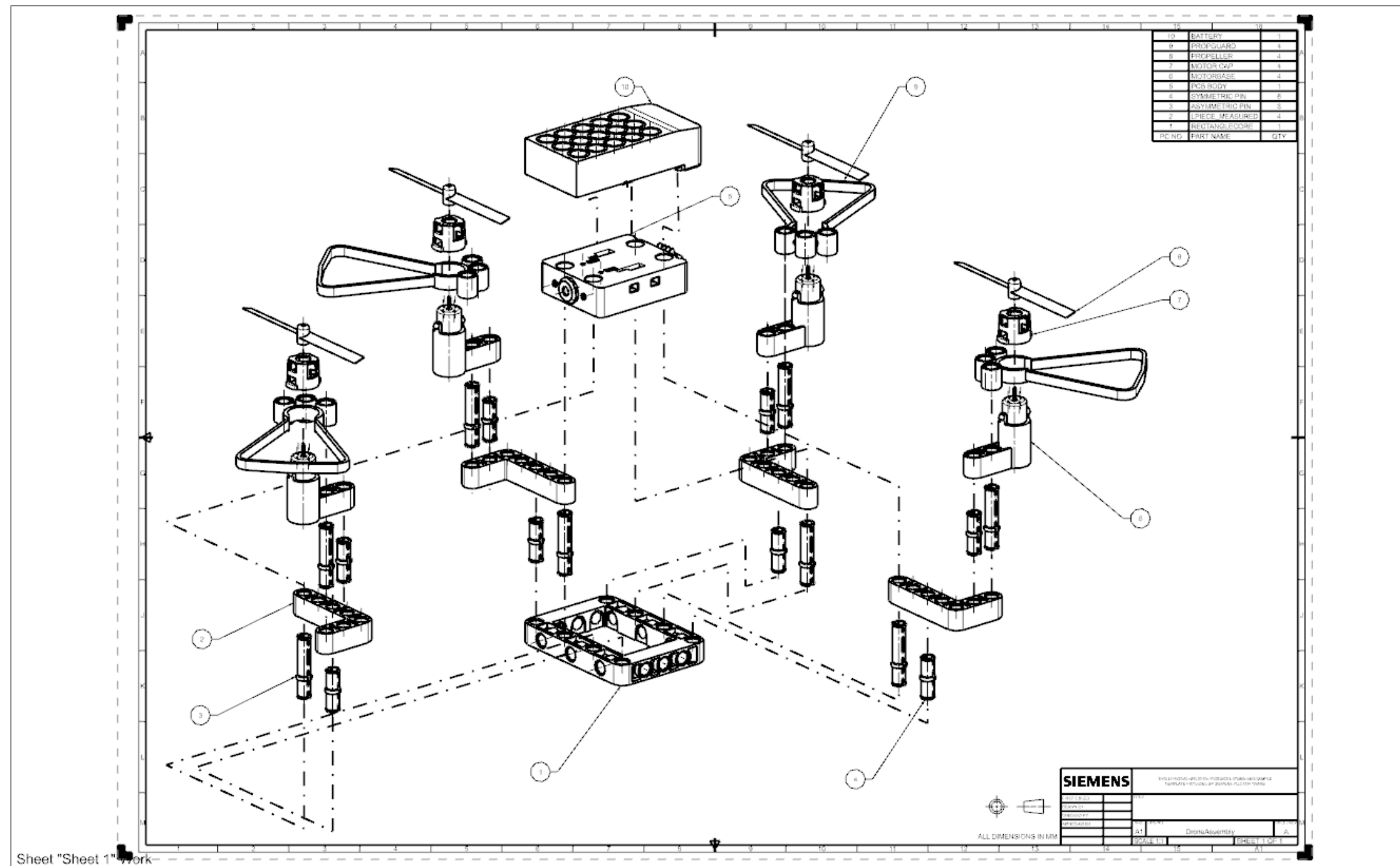
Full Assembly



Exploded Assembly



Bill of Materials

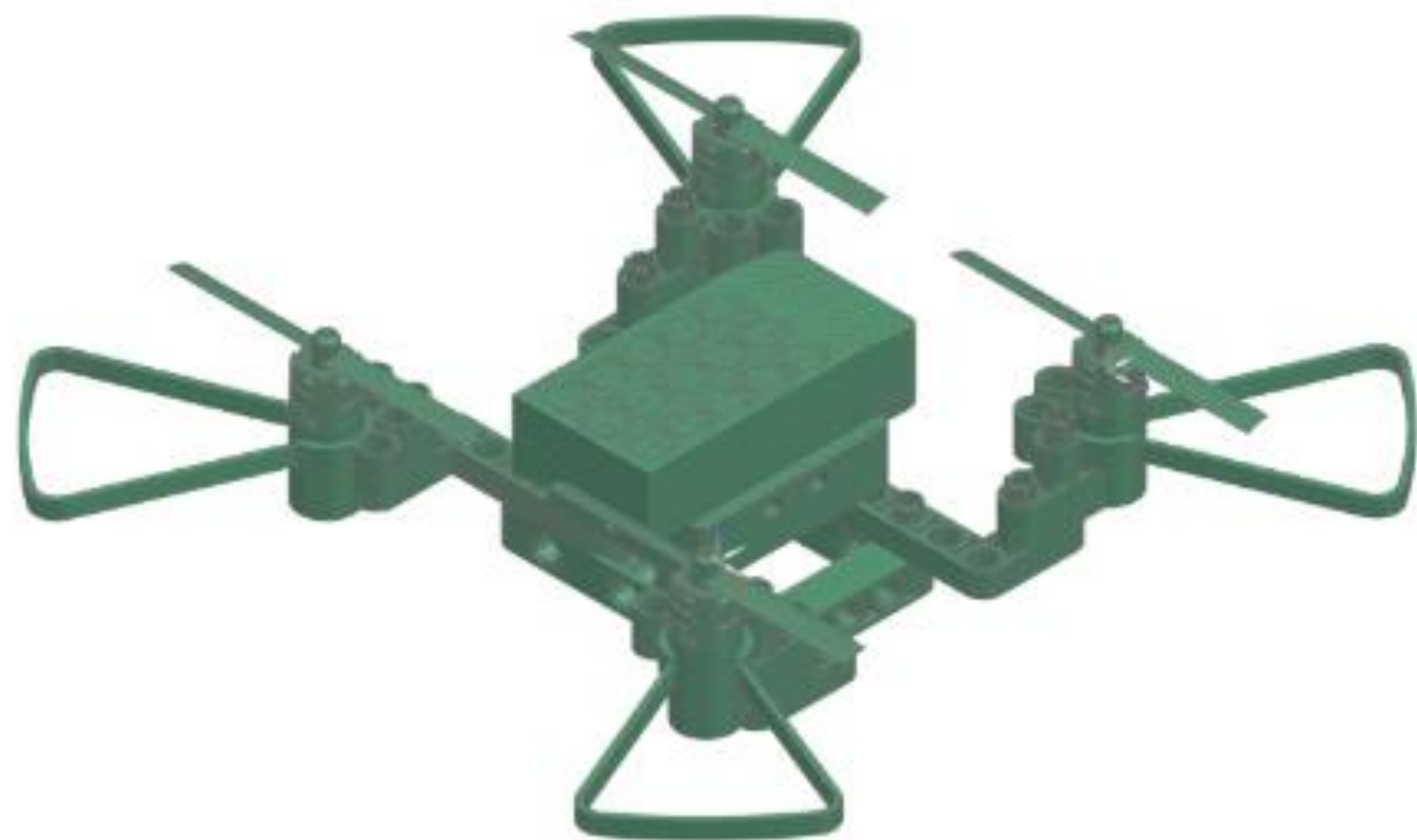


Assembly Mass Information

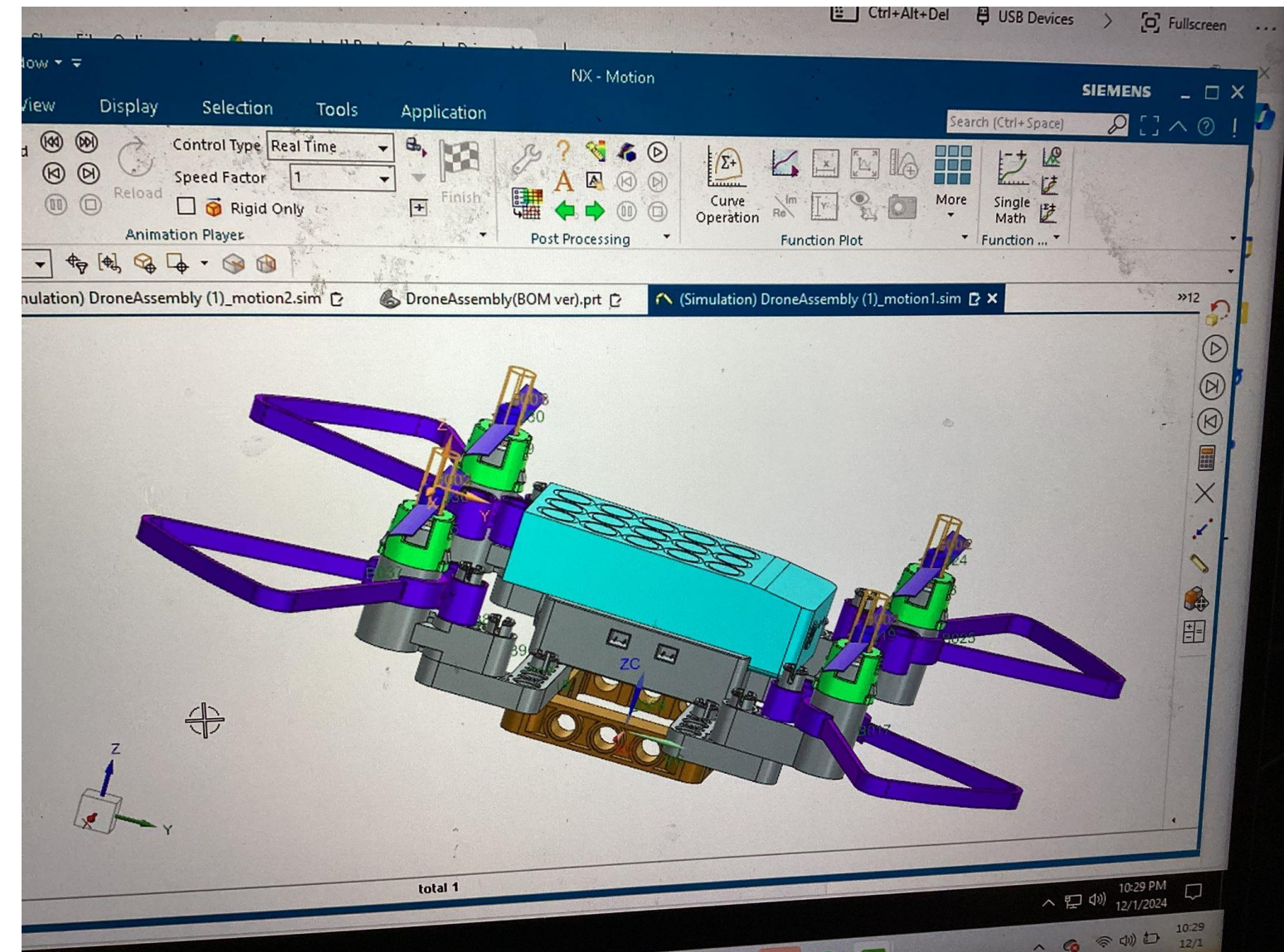
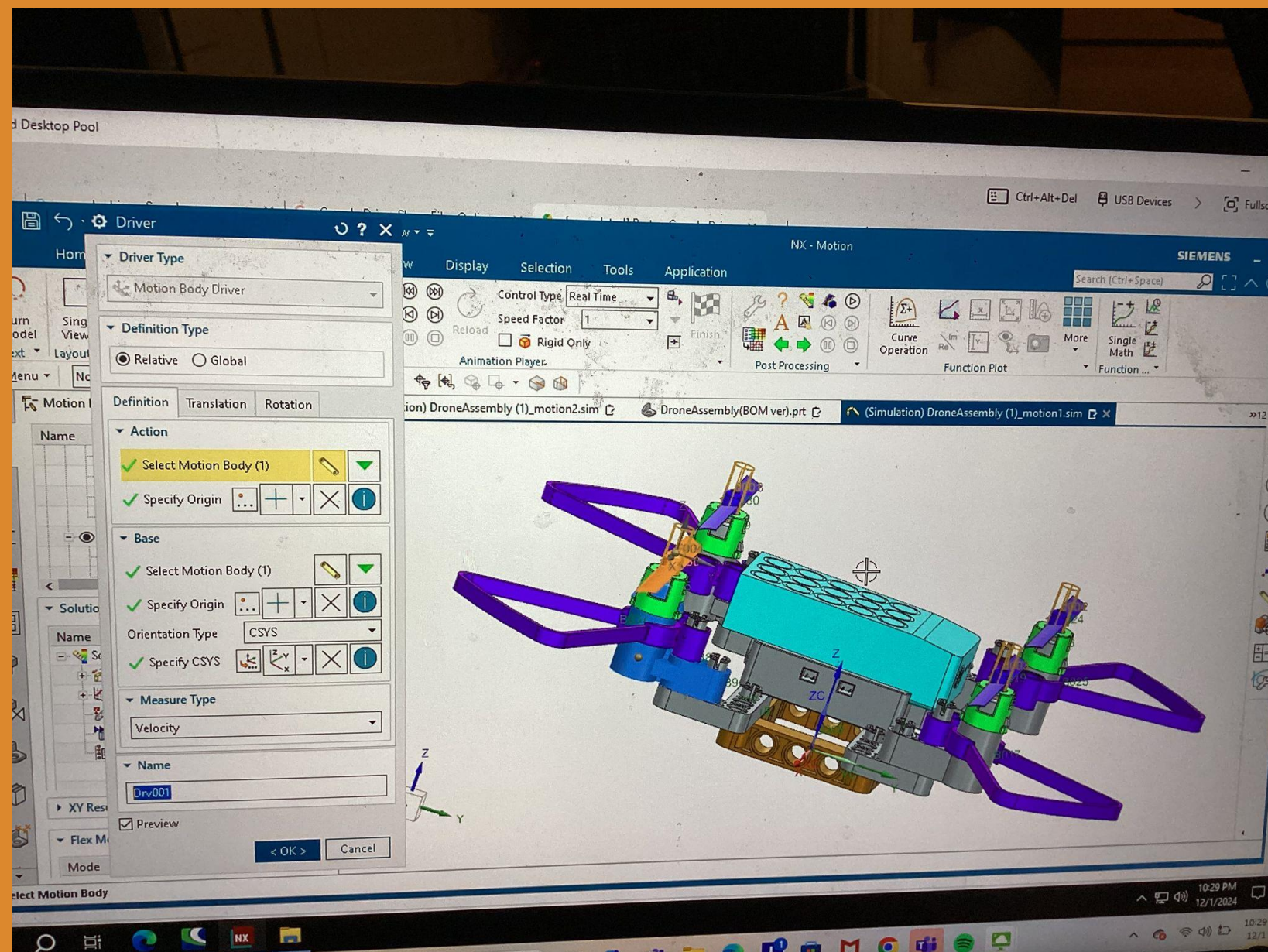
[illegible]

FEA Analysis

Force	Force magnitude	Force direction	Point of application
Aerodynamic drag	7.35N on each propeller	Against motion (rearwards)	Propellers
Thrust force	1.5N on each motor	Upwards (against gravity)	Motor mounts
Inertial force	40N	Center of mass	Outward (centripetal)
Gravitational force	5N	Center of mass	downwards
Wind gust	5.51N	Against the direction of motion	The side surface of the drone



Motion Analysis



Thank You!

